Research Proposal: The Effect of Spaced Repetition on Vocabulary Retention in Adult Second-Language Learners

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Abstract

Learners forget a large share of newly studied second-language words within weeks. Laboratory research shows that spacing practice over time improves retention, but classroom studies with adult learners have used short retention intervals and have not compared equal and expanding schedules directly. This project proposes a pre-registered experiment in which 120 adult learners study 60 unfamiliar nouns under a massed, an equal-spaced or an expanding-spaced schedule, with retention tested one and four weeks later.

1 Introduction and Background

Knowing a word means being able to retrieve its form and meaning long after the lesson in which it was first met (Nation, 2013). The forgetting curve measured by Ebbinghaus (1885) shows that most of what is learned is lost within days unless it is practised again. The spacing effect, in which study distributed over time produces better long-term retention than the same amount of study in one session, is among the most robust findings in memory research (Cepeda et al., 2006). It remains unclear, however, how spacing should be scheduled for second-language vocabulary: comparisons of expanding and equal schedules have produced mixed results, and most of them measured retention after a week or less (Nakata, 2015). This proposal addresses that gap with a four-week retention interval and enrolled adult learners.

2 Research Questions and Hypotheses

RQ1 Does spacing retrieval practice over several days improve retention of new vocabulary at one and four weeks, compared with massed practice of the same duration? **H1:** Both spaced conditions will outperform the massed condition at both delays.

RQ2 Does an expanding schedule produce better retention than an equal-interval schedule? **H2:** Expanding spacing will yield higher cued recall than equal spacing at four weeks, with no reliable difference at one week.

RQ3 Is the benefit of spacing moderated by prior proficiency? **H3:** The advantage of spaced over massed practice will be larger for lower-proficiency learners.

3 Literature Review

The meta-analysis by Cepeda et al. (2006) of 317 experiments on distributed practice found that spaced study improved verbal recall in almost every comparison and that the optimal gap between sessions grows with the retention interval. Karpicke and Roediger (2008) showed that repeated testing of foreign-language word pairs produced about 80 % recall after one week, against roughly 35 % for repeated study alone. Within second-language research, Nation (2013) argues that vocabulary programmes should plan repeated retrieval of each word, but the studies that have tested specific schedules used few items and short retention intervals: Nakata (2015) found only a limited advantage for expanding over equal spacing with 20 word pairs and a one-week delayed test. No study has combined a four-week retention interval with a direct comparison of all three schedules.

4 Methodology

4.1 Design

The study is a pre-registered, between-subjects experiment with three conditions: *massed*, *equal-spaced* and *expanding-spaced* retrieval practice. Each participant completes six retrieval attempts per word: all six in one 60-minute session (massed), two in each of three 20-minute sessions on days 1, 4 and 7 (equal-spaced), or on days 1, 2 and 7 (expanding-spaced). Retention is tested one and four weeks after the last session.

4.2 Participants

A power analysis for the three-group comparison that underlies H1 (G*Power 3.1, one-way ANOVA; medium-to-large effect size $f = 0.30$, $\alpha = 0.05$, power $= 0.80$) requires 111 participants. We will recruit 120 adult learners (40 per condition) from beginner and lower-intermediate courses at the university language centre to allow for attrition. Ethics approval will precede recruitment and participants will give written consent.

4.3 Materials and Procedure

Sixty concrete nouns from a language unfamiliar to all participants (Finnish) are paired with English translations. Session 1 begins with a 40-item proficiency test and one study exposure to all 60 words. In each retrieval attempt the English meaning is shown, the learner types the Finnish form and receives the correct answer as feedback. The delayed tests consist of cued recall of the form from the meaning and a four-option meaning recognition test, run in the same browser-based training platform as the practice sessions.

4.4 Data Analysis

Item-level accuracy will be analysed with mixed-effects logistic regression, with condition, delay and their interaction as fixed effects, proficiency as a continuous moderator, and random intercepts for participants and items. The hypotheses and analysis script will be pre-registered on the Open Science Framework before data collection begins.

4.5 Timeline

Table 1 shows the planned twelve-month schedule.

Table 1: Project timeline; filled cells mark the active quarters.

Phase	Activity	Months 1–3	4–6	7–9	10–12
1	Ethics approval, stimuli and pilot study	■			
2	Recruitment and training sessions		■	■	
3	One- and four-week retention tests			■	■
4	Analysis, thesis writing and dissemination				■

5 Expected Outcomes and Significance

We expect both spaced conditions to retain substantially more words than the massed condition at four weeks, with a small further advantage for the expanding schedule. If H3 is supported, lower-proficiency learners gain the most from spaced retrieval. The project will produce a master's thesis, a journal article, an open dataset and a review schedule that teachers and spaced-repetition software can adopt directly.

6 Budget

Table 2 summarises the requested funding.

Table 2: Requested budget.

Item	Amount (USD)
Participant compensation (120 participants × \$40)	4,800
Research assistant (200 hours × \$20)	4,000
Training platform licence (12 months)	600
Conference travel and dissemination	1,500
Open-access publication fee	2,000
Total	**12,900**

References

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